Seyed Kiarash Eshraghi Naeini

B.Sc. Student in Electrical Engineering, Sharif University of Technology, Tehran, Iran
8kia.eshraghi4@gmail.com · kiarash.eshraghi84@sharif.edu
seyedkiarash.github.io · github.com/seyedkiarash · linkedin.com/in/seyed-kiarash-eshraghi-6b16a3447

RESEARCH INTERESTS

Speech and audio signal processing, voice conversion, estimation theory and optimization, graph signal processing, convolutional and graph neural networks for audio, compressed sensing.

EDUCATION

B.Sc. in Electrical Engineering, Sharif University of Technology 2024 – 2028 (expected)
GPA: **18.59 / 20** · Ranked **22nd of 208** in the entering class (department average GPA 16.51)
Selected courses: Signals & Systems (19.4), Engineering Probability & Statistics (19.4), Mathematical Methods in Engineering (19.7), Engineering Mathematics (18.6), Electrical Circuit Theory (20), Electromagnetics (20), Electronics 1 (19.5), Fundamentals of Programming (20), Object-Oriented Programming (19.9)
In progress (Fall 2026): **Graph Signal Processing** (graduate course), Communication Systems, Linear Control Systems

RESEARCH EXPERIENCE

Research Collaborator, Multimedia Lab, Department of Electrical Engineering, Sharif University of Technology Sep 2026 – Present
Supervisor: Dr. Arash Amini

- Working on **voice conversion**: currently developing non-generative approaches grounded in estimation theory and optimization.
- Next step: extending the work to generative voice conversion built on the same estimation-theoretic and optimization framework.
- Longer term: broadening to other speech and audio processing methods beyond voice conversion, with particular interest in CNN- and GNN-based approaches.

SELECTED PROJECTS

Separating Speech from a Siren Using Notch Filters — *Signals & Systems, Dr. Arash Amini* 2026

- Designed a 16th-order digital notch filter by pole–zero placement in the z-plane (four cascaded 4th-order sections for numerical stability), with frequency-dependent pole radii to protect voice formants; strongly attenuated the siren with only slight muffling of the voice. *MATLAB*

Active Filters and Analog Noise Cancellation — *Circuit Theory, Dr. Rouhollah Amiri* Fall 2025

- Designed a state-variable filter (f_c = 20 kHz, Q = 0.75) and an analog noise-cancelling chain with a 2nd-order all-pass delay stage. The full chain raises the SNR of the recording from 8.4 dB to 41.6 dB. In a later review, found that the delay stage was not truly all-pass and derived corrected component values that make it a true all-pass with the intended 45 µs delay. *LTspice, Simulink*

3D Electromagnetic Field Visualizer for Waveguides and Optical Fibers — *Fields & Waves, Dr. Mohammad Memarian* 2026

- MATLAB tool that renders animated 3D E- and H-field distributions of TE/TM modes in parallel-plate, rectangular and circular waveguides and optical fibers, with cutoff and mode-validity checks. *MATLAB*

Multi-Armed Bandit: Theory and Simulation — *Probability & Statistics, Dr. Mohammad Hossein Yassaee Meybodi* Fall 2025

- Regret analysis, Hoeffding bounds and UCB, Beta–Bernoulli updates and Thompson sampling; corrected unfair algorithm comparisons and re-ran all simulations with confidence bands over hundreds of runs. *Python*

Logic Circuits Lab on FPGA — *Dr. Mohammad Reza Movahhedin* Fall 2025

- 32 Verilog designs on a Zynq-7010 board (Hamming/SECDED codes, BCD arithmetic, multiplexed displays, stopwatch, games), each verified by a self-checking testbench. *Verilog, Vivado*

Circuit Simulator in C++ — *Object-Oriented Programming (team), Dr. Abdollah Arasteh* Spring 2025

- SPICE-like simulator using modified nodal analysis (transient, DC, AC) with a Qt GUI; results validated against ngspice. *C++, Eigen, Qt*

Computational Assignments — *Mathematical Methods in Engineering; Signals & Systems* 2026

- SVD image compression, PageRank, constrained optimization (KKT, penalty and augmented Lagrangian methods), linear-algebra algorithms from scratch; Z-transform, Fourier analysis and image filtering. *Python, MATLAB*

TEACHING EXPERIENCE

Head Teaching Assistant, Circuit Theory — Dr. Rouhollah Amiri *(designed and graded assignments, managed the TA team; 40 students)* Fall 2026

Teaching Assistant, Probability & Statistics — Dr. Mohammad Hossein Yassaee Meybodi *(designed assignments; 60 students)* Fall 2026

Head TA (Assignments), Object-Oriented Programming — Dr. Abdollah Arasteh *(managed assignment TAs; 105 students)* Spring 2026

Teaching Assistant, Electromagnetics — Dr. Mohammad Memarian *(problem-solving sessions; 170 students)* Spring 2026

Teaching Assistant, Fundamentals of Programming — Dr. Abdollah Arasteh *(designed assignments; 210 students)* Fall 2025

HONORS

Ranked **198th**, Iranian National University Entrance Exam (Mathematics & Physics) 2024

Bronze Medal, Iranian National Physics Olympiad *(national summer camp, top 40)* 2023

SKILLS

Programming	Python (NumPy, SciPy, Matplotlib, Jupyter), C++, MATLAB / Simulink, Verilog
Tools	LTspice, Vivado, Icarus Verilog, Qt, Git, LaTeX
Languages	Persian (native), English (upper-intermediate), Korean (beginner)